

ISE 5405: Optimization I

Simplex II: Directions, Reduced Costs, and Column Geometry

BT §§3.1–3.2 and §3.6

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What You Should Be Able to Do

1. Distinguish adjacency of points from adjacency of bases.
2. Construct a basic direction from a nonbasic column.
3. Decide whether that direction permits a positive feasible step.
4. Derive and interpret a reduced cost.
5. Test whether a feasible basis is optimal and interpret the column geometry.

Retrieval: What Does a Basis Give Us?

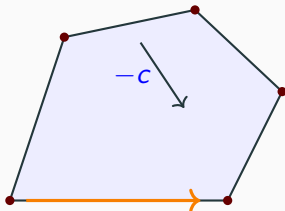
For $P = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\}$, with $\text{rank}(A) = m$, a basis contains m independent columns:

$$x_B = A_B^{-1}b, \quad x_N = 0.$$

Two separate questions

Is $x_B \geq 0$? If so, can an adjacent basis improve the objective?

Edges Suggest an Algorithm



Simplex idea

Start at a corner, identify an improving incident edge, and move to the next corner. Algebra must give both the direction and the maximum step.

Adjacent Basic Feasible Solutions

Two distinct vertices of a polyhedron in $\mathbb{R}^n$ are **adjacent** when they share $n - 1$ linearly independent active constraints. In standard form, these vertex points are basic feasible solutions.

For a nondegenerate standard-form BFS:

- the bases differ by one column;
- one nonbasic variable enters and one basic variable leaves; and
- the feasible points are joined by an edge.

Adjacent Bases Need Not Mean Distinct Points

At a degenerate BFS, at least one basic variable is zero.

- Several bases can encode the same feasible point.
- Exchanging one column may produce a zero-length step.
- Adjacent bases and adjacent vertices are not interchangeable without a nondegeneracy qualification.

Consequence

Simplex operates on bases; its geometric location need not change at every pivot.

Quick Check: Are These Points Adjacent?

Let

$$P = \{(x_1, x_2) : x_1 + x_2 \leq 4, x_1 \leq 3, x_2 \leq 2, x \geq 0\}.$$

1. List active constraints at $(0, 0)$, $(3, 0)$, and $(3, 1)$.
2. Which pairs share one independent active constraint?
3. Which pairs are joined by an edge?

Work for two minutes

Use the definition, not only a mental sketch.

Quick Check: Solution

Point	Active constraints
$(0, 0)$	$x_1 = 0, x_2 = 0$
$(3, 0)$	$x_1 = 3, x_2 = 0$
$(3, 1)$	$x_1 = 3, x_1 + x_2 = 4$

$(0, 0)$ – $(3, 0)$ share $x_2 = 0$; $(3, 0)$ – $(3, 1)$ share $x_1 = 3$. Those pairs are adjacent. The first and third points are not.

The Standard-Form Setting

We study

$$\min\{c^\top x : Ax = b, x \geq 0\}.$$

Partition according to a feasible basis:

$$A = [A_B \ A_N], \quad x = \begin{bmatrix} x_B \\ x_N \end{bmatrix} = \begin{bmatrix} A_B^{-1}b \\ 0 \end{bmatrix}.$$

Write $B = A_B$ when there is no ambiguity.

Definition

d is a feasible direction at $x \in P$ if some $\epsilon > 0$ satisfies $x + \theta d \in P$ for every $0 \leq \theta \leq \epsilon$.

- Equalities require $Ad = 0$.
- Nonnegativity requires $x_i + \theta d_i \geq 0$.
- Both conditions matter.

Preserving the Equalities

If $Ax = b$, then

$$A(x + \theta d) = b + \theta Ad.$$

Thus the line preserves the equations precisely when

$$Ad = 0.$$

Null-space view

Every edge direction in standard form lies in the null space of A .

$$x_i + \theta d_i \geq 0.$$

- When $x_i > 0$, a negative d_i is allowed for a small step.
- When $x_i = 0$, a positive step requires $d_i \geq 0$.

Degeneracy warning

$Ad = 0$ alone does not make d feasible at x . A zero basic component can block a proposed direction immediately.

Choose One Nonbasic Variable

For a nonbasic index j , prescribe

$$d_j = 1, \quad d_k = 0 \quad (k \in N \setminus \{j\}).$$

The basic components must compensate:

$$A_B d_B + A_j = 0.$$

This choice defines the j -th **basic direction**.

Solve for the Basic Components

Since $B = A_B$ is nonsingular,

$$Bd_B = -A_j \implies \boxed{d_B = -B^{-1}A_j.}$$

Define

$$u := B^{-1}A_j, \quad d_B = -u.$$

One linear solve produces the direction.

Check the Direction Algebra

$$\begin{aligned}Ad &= A_B d_B + A_j d_j \\ &= B(-B^{-1}A_j) + A_j \\ &= -A_j + A_j = 0.\end{aligned}$$

Every basic direction preserves the equality system. Feasibility still requires checking the signs of components that are zero at the current BFS.

Does the Direction Permit a Step?

$$x_B(\theta) = x_B - \theta u.$$

Therefore

$$\theta^* = \min_{i: u_i > 0} \frac{(x_B)_i}{u_i},$$

provided some $u_i > 0$.

- $\theta^* > 0$: a genuine edge step exists.
- $\theta^* = 0$: the direction is blocked, though a basis pivot is possible.

Running Example

$$\begin{array}{ll}\min & -10x_1 - 12x_2 - 12x_3 \\ \text{s.t.} & x_1 + 2x_2 + 2x_3 + x_4 = 20, \\ & 2x_1 + x_2 + 2x_3 + x_5 = 20, \\ & 2x_1 + 2x_2 + x_3 + x_6 = 20, \quad x \geq 0.\end{array}$$

This model will connect matrix simplex, tableaus, and revised simplex.

Initial Basis and Basic Feasible Solution

Choose $B = [A_4 \ A_5 \ A_6] = I$. Then

$$x_B = (20, 20, 20)^T, \quad x_N = (x_1, x_2, x_3) = 0.$$

Thus

$$x^{(0)} = (0, 0, 0, 20, 20, 20)$$

is a nondegenerate BFS with objective value 0.

Your Turn: Direction for x_1

At the slack basis, let x_1 increase.

1. Compute $u = B^{-1}A_1$.
2. Form all six components of d .
3. Verify $Ad = 0$.
4. Identify the variables that decrease.

Work for three minutes

Keep the variable order $(x_1, \dots, x_6)$.

Direction for x_1 : Solution

$$u = A_1 = (1, 2, 2)^\top, \quad d_B = (-1, -2, -2)^\top.$$

With $d_1 = 1$,

$$d = (1, 0, 0, -1, -2, -2)^\top.$$

Indeed, $Ad = A_1 - A_4 - 2A_5 - 2A_6 = 0$. All three slack variables decrease.

How Far Along That Edge?

$$x_B(\theta) = \begin{bmatrix} 20 \\ 20 \\ 20 \end{bmatrix} - \theta \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

$$\theta^* = \min \left\{ \frac{20}{1}, \frac{20}{2}, \frac{20}{2} \right\} = 10.$$

Both x_5 and x_6 reach zero. The ratio tie makes the new point degenerate.

Objective Slope Along a Direction

$$c^T(x + \theta d) = c^T x + \theta c^T d.$$

For the j -th basic direction,

$$c^T d = c_j + c_B^T d_B = c_j - c_B^T B^{-1} A_j.$$

The coefficient of θ is the objective slope along that direction.

Definition

The reduced cost of x_j relative to B is

$$\bar{c}_j = c_j - c_B^\top B^{-1} A_j.$$

Since $c^\top d = \bar{c}_j$, a negative reduced cost gives an improving slope for a minimization problem.

Direct Cost Minus Displacement Cost

$$\bar{c}_j = \underbrace{c_j}_{\text{direct cost}} - \underbrace{c_B^\top B^{-1} A_j}_{\text{cost of displaced basics}} .$$

- A basic variable has zero reduced cost.
- Reduced cost depends on the current basis.
- It is a slope; the ratio test separately determines the available step.

Reduced Costs at the Slack Basis

$c_B = 0$ and $B = I$, so $\bar{c}_j = c_j$:

	x_1	x_2	x_3	x_4	x_5	x_6
$\bar{c}_j$	-10	-12	-12	0	0	0

All three structural variables offer an improving slope.

Choose an Entering Variable

For this worked path, choose the eligible variable x_1 . This is a deliberate pivot choice, not the most-negative-cost rule.

1. Why is x_1 eligible?
2. Which variables would most-negative pricing consider instead?
3. Why can two valid pivot rules trace different paths?

Thirty-second check

Separate eligibility from the chosen pivot rule.

Entering Choice: Solution

$$\bar{c}_1 = -10, \quad \bar{c}_2 = -12, \quad \bar{c}_3 = -12.$$

- All three structural variables are eligible because their reduced costs are negative.
- The worked path selects x_1 .
- Most-negative pricing would instead choose between x_2 and x_3 .

The First Edge Step

For entering x_1 ,

$$u = (1, 2, 2)^{\top}, \quad \theta^* = 10.$$

Use the first tied minimum-ratio row, so x_5 leaves. Then

$$B_1 = [A_4 \ A_1 \ A_6],$$

$$x^{(1)} = (10, 0, 0, 10, 0, 0), \quad c^{\top} x^{(1)} = -100.$$

One Edge, Two Newly Active Constraints

$$x^{(1)} = x^{(0)} + 10(1, 0, 0, -1, -2, -2).$$

- x_5 reaches zero and leaves.
- x_6 also reaches zero because the ratio test tied.
- x_6 remains basic at zero, so $x^{(1)}$ is degenerate.

The New Basis Inverse

For $B_1 = [A_4 \ A_1 \ A_6]$,

$$B_1^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & -1 & 1 \end{bmatrix}, \quad B_1^{-1}b = \begin{bmatrix} 10 \\ 10 \\ 0 \end{bmatrix}.$$

The zero basic component records degeneracy algebraically.

Reduced Costs at the New Basis

With $c_B = (0, -10, 0)^\top$,

$$c_B^\top B_1^{-1} = (0, -5, 0), \quad \bar{c} = (0, -7, -2, 0, 5, 0).$$

Both nonbasic x_2 and x_3 have negative reduced cost.

Two Improving Slopes at a Degenerate BFS

For x_2 ,

$$u = B_1^{-1}A_2 = \begin{bmatrix} \frac{3}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}, \quad \bar{c}_2 = -7.$$

The zero basic value of x_6 blocks this direction at step zero. For x_3 , however,

$$B_1^{-1}A_3 = (1, 1, -1)^\top, \quad \bar{c}_3 = -2.$$

The Pivot Choice Can Move or Stall

$$\theta^*(x_2) = \min \left\{ \frac{10}{3/2}, \frac{10}{1/2}, \frac{0}{1} \right\} = 0,$$

$$\theta^*(x_3) = \min \left\{ \frac{10}{1}, \frac{10}{1} \right\} = 10.$$

- Choosing x_2 causes a degenerate basis change at point D .
- The worked path chooses x_3 , lets x_4 leave, and moves to $B = (0, 0, 10)$.
- Both choices are eligible; the basis path records more than the visible edge path.

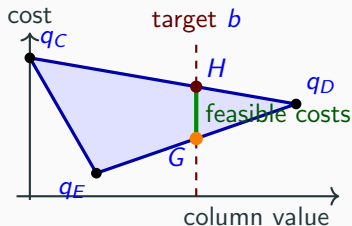
What a Negative Reduced Cost Really Says

For minimization, $\bar{c}_j < 0$ says the objective decreases along the j -th basic direction *if a positive feasible step is available*. It does not guarantee:

- a positive step at a degenerate BFS;
- the largest total improvement; or
- that the pivot reaches a distinct point.

Column Geometry: A Literal 2D Picture

General: $q_j = (A_j, c_j) \in \mathbb{R}^{m+1}$. Here $m = 1$, so $q_j = (a_j, c_j) \in \mathbb{R}^2$ lies on the page—this is not a perspective drawing. With $\mathbf{1}^\top x = 1$, $(b, z) = \sum_j x_j q_j$, $x \geq 0$, is a convex combination.



Exact example

$q_C = (0, 3)$, $q_D = (4, 2)$, and $q_E = (1, \frac{1}{2})$; target $b = \frac{5}{2}$.

Vertical fiber

Fix $a^\top x = b$: take the vertical slice of $\text{conv}\{q_C, q_D, q_E\}$ above b .

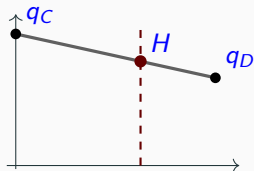
Optimization

The lowest feasible point is $G = (\frac{5}{2}, \frac{5}{4})$, so its height is the minimum cost.

One Pivot, Shown in Three Separate Views

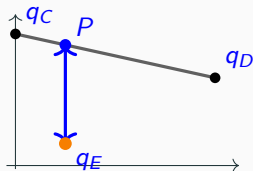
All three panels use the same literal 2D axes: horizontal column value and vertical cost. In higher dimensions, segments become simplices.

1. Current basis



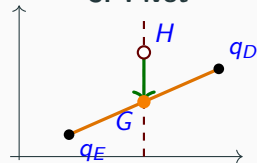
CD meets the fiber at
 $H = (\frac{5}{2}, \frac{19}{8})$.

2. Price E



$\bar{c}_E = \frac{1}{2} - \frac{11}{4} = -\frac{9}{4}$; the
 upward cost gap is
 $-\bar{c}_E$.

3. Pivot



E enters, C leaves, and
 DE meets the fiber at
 $G = (\frac{5}{2}, \frac{5}{4})$.

Reduced-cost gap versus objective decrease

$$\theta^* = \frac{1}{2}, \quad z_G - z_H = \theta^* \bar{c}_E = \frac{1}{2} \left(-\frac{9}{4} \right) = -\frac{9}{8}.$$

Why Nonnegative Reduced Costs Prove Optimality

For any feasible x ,

$$x_B = B^{-1}b - B^{-1}A_N x_N.$$

Therefore

$$c^T x = c_B^T B^{-1}b + \bar{c}_N^T x_N.$$

If $\bar{c}_N \geq 0$ and $x_N \geq 0$, no feasible solution has value below the current basic value.

The Two Checks for an Optimal Basis

$$\boxed{B^{-1}b \geq 0} \quad \text{and} \quad \boxed{\bar{c}_N = c_N - A_N^T B^{-T} c_B \geq 0}.$$

Check	Meaning
$B^{-1}b \geq 0$	primal feasibility
$\bar{c}_N \geq 0$	no improving nonbasic slope

The Converse Needs Care

- A feasible basis with nonnegative reduced costs is optimal.
- At a nondegenerate optimal BFS, a negative reduced cost would provide a positive improving step.
- At a degenerate optimum, one basis can have a negative reduced cost whose direction is blocked at step zero.
- The point can possess another basis with nonnegative reduced costs.

Final Check: A Candidate Optimal Basis

For the running example, take

$$B_* = [A_2 \ A_3 \ A_1], \quad x_{B_*} = (4, 4, 4)^\top,$$

and suppose

$$c_{B_*}^\top B_*^{-1} = \left(-\frac{18}{5}, -\frac{8}{5}, -\frac{8}{5}\right).$$

1. Verify feasibility.
2. Compute reduced costs of x_4, x_5, x_6 .
3. State the solution and value.

Candidate Basis: Solution

The basic variables $(x_2, x_3, x_1) = (4, 4, 4)$ give

$$x^* = (4, 4, 4, 0, 0, 0), \quad c^\top x^* = -136.$$

The slack columns are unit vectors, so

$$\bar{c}_4 = \frac{18}{5}, \quad \bar{c}_5 = \frac{8}{5}, \quad \bar{c}_6 = \frac{8}{5}.$$

Basic values and nonbasic reduced costs are nonnegative; B_* is optimal.

What We Built Today

1. $u = B^{-1}A_j$ and $d_B = -u$ define a basic direction.
2. The ratio test decides whether a positive step is possible.
3. $\bar{c}_j = c_j - c_B^\top B^{-1}A_j$ is its objective slope.
4. Feasible basic values and nonnegative reduced costs certify optimality.
5. Degeneracy separates a basis change from a geometric move.

Next

We turn these ingredients into a complete primal-simplex iteration.